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Poisson GLM

Objective

The aim of this analysis is to model the number of high-consumption days per household-month ($n_{high\ days}$). This variable represents count data, taking non-negative integer values. A plain linear model is not appropriate here, as it assumes normally distributed errors and can produce negative predictions — both of which are incompatible with count outcomes.

A Poisson Generalised Linear Model (GLM) is used instead, as it is specifically designed for non-negative integer outcomes and naturally allows the variance to grow with the mean. By including an offset for the number of observed days, the model estimates rates, allowing interpretation of results in terms of relative changes (rate ratios).

Model Setup

```
## n_high_days ~ heating_degree_days + temp_avg + living_area +      building_type + heatpump_type + ha

## glm(formula = form, family = poisson(link = "log"), data = dat_count,
##      offset = log_n_days)
```

The response variable is the number of high-consumption days per household-month ($n_{high\ days}$). Because the number of observed days can vary slightly across household-months, the model includes the logarithm of the observed days ($\log(n_{days})$) as an offset. This adjustment effectively converts the outcome into a rate — high-consumption days per observed day — so that observations remain comparable even when exposure differs between months.

To model these rates, a Poisson GLM with a log-link is fitted:

$$\log(\mathbb{E}[Y_i]) = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_p X_{pi} + \log(n_{days_i})$$

The log-link guarantees positive fitted values and supports a multiplicative interpretation of effects.

Model Results

The results are interpreted in terms of *percentage changes in the expected rate*, holding all other variables constant.

Building type (house vs flat): Households living in houses have a rate of high-consumption days approximately 363% higher than those living in flats $((4.632 - 1) \times 100\%)$. This is by far the strongest effect in the model, clearly visible in the boxplot, and indicates that building type is the dominant structural driver of consumption patterns.

Average temperature: The rate ratio of 0.915 implies that each additional degree Celsius is associated with an 8.5% decrease in the expected rate of high-consumption days $((1 - 0.915) \times 100\%)$. This reflects reduced heating needs at warmer temperatures, consistent with the strong negative trend visible in the scatterplot.

Heating degree days: The rate ratio is marginally below 1 (≈ 0.999) and not statistically significant. Once average temperature is already in the model, heating degree days add little additional explanatory power, as both variables capture overlapping thermal information. This collinearity explains why the coefficient is effectively zero.

Number of residents: Each additional household member increases the expected rate by approximately 8.9% $((1.089 - 1) \times 100\%)$. This reflects the intuitive result that larger households consume more energy overall.

Living area: The rate ratio of 1.001 indicates a statistically significant but practically small positive effect — each additional square metre increases the expected rate by about 0.1%. The effect accumulates for larger dwellings but remains modest compared to building type or household size.

Ground-source heat pump: Households equipped with a ground-source heat pump show a rate approximately 9.3% lower than the reference heat pump type $((1 - 0.907) \times 100\%)$. This is consistent with ground-source systems being more energy-efficient, particularly during colder periods.

Floor heating: Households with floor heating show a rate approximately 13% higher than those without $((1.130 - 1) \times 100\%)$. This likely reflects the greater energy input required by radiant floor systems to maintain comfort temperatures.

Weekend share: A higher proportion of weekend days within a month is associated with a substantially lower rate of high-consumption days ($RR \approx 0.665$). However, this effect has a very wide confidence interval and becomes non-significant once overdispersion is accounted for, suggesting considerable uncertainty in this estimate.

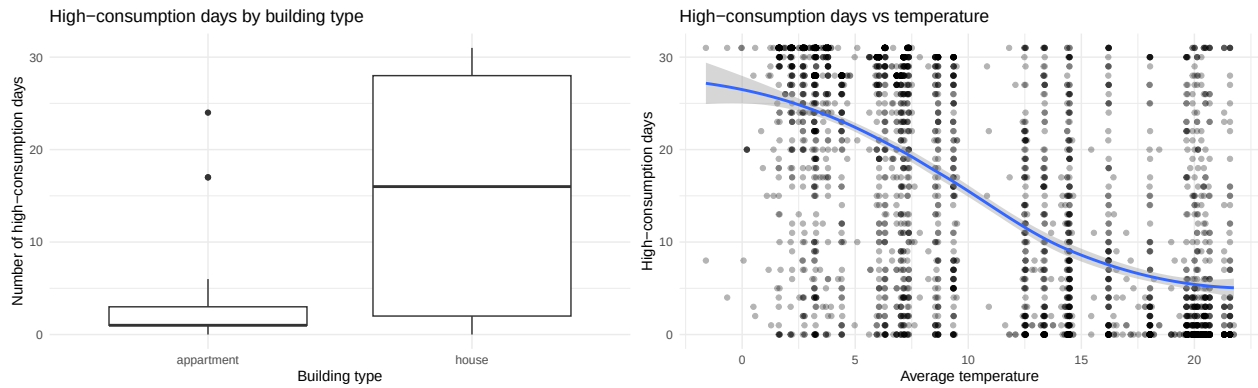
Overall, structural and household characteristics (building type, size, and number of residents) exhibit the strongest and most reliable effects, while temperature captures an important environmental influence.

Visual Evidence

To complement the model output, the plots below provide direct visual support for the main relationships in the data.

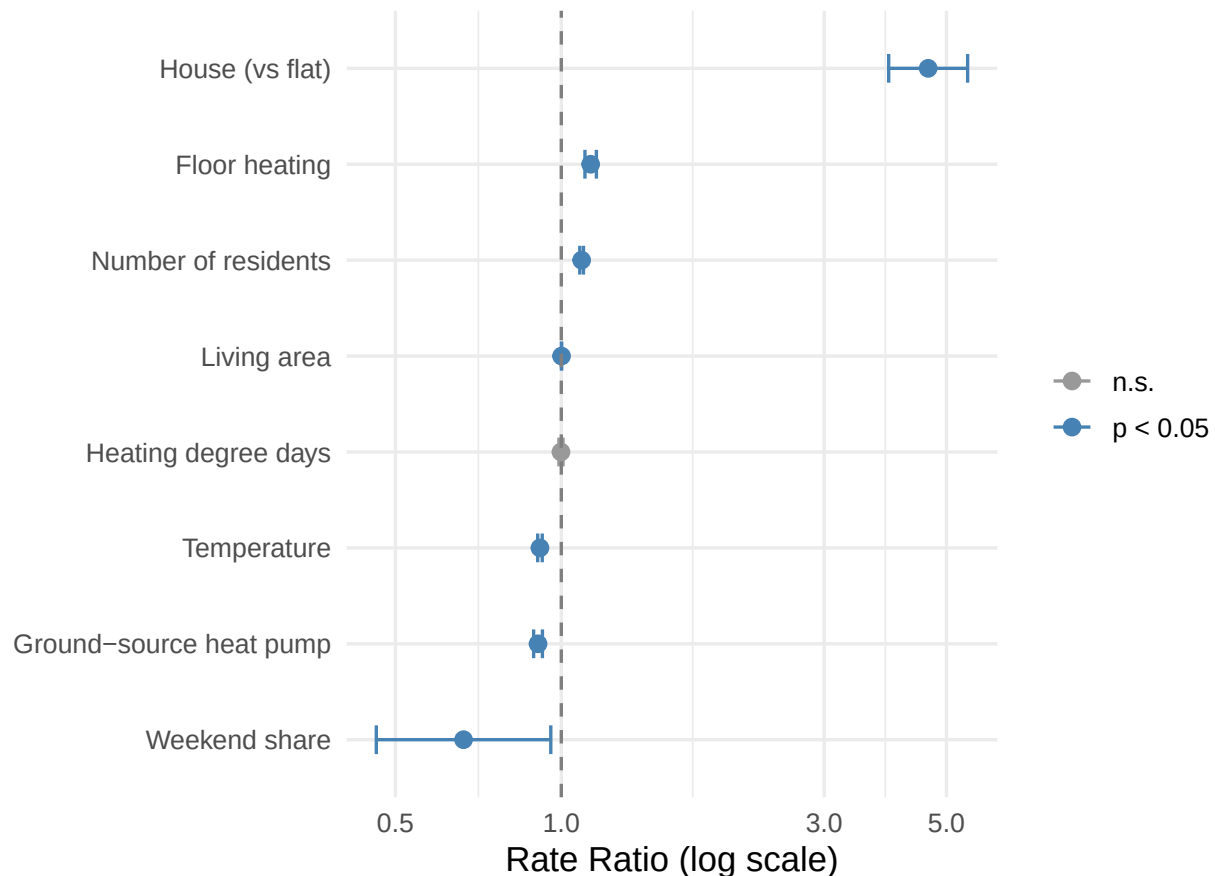
The *boxplot by building type* shows a striking separation between apartments and houses. Apartments cluster tightly with a median close to 3 high-consumption days and only two mild outliers, while houses display a median around 16 days with a very wide interquartile range extending from nearly 0 to 28. This large difference in both level and spread confirms that building type is the dominant predictor in the model.

The *scatterplot of high-consumption days versus average temperature* reveals a strong and consistent negative pattern. At temperatures near 0°C, many households record close to 30 high-consumption days, whereas above 15°C the counts concentrate near zero. The smooth LOESS curve decreases steeply in the 0–10°C range and flattens at higher temperatures, suggesting diminishing marginal effects of warmth once heating demand is already low.



The *rate-ratio plot* below summarises all model effects on a log scale. *House (vs flat)* stands far to the right of the reference line ($RR \approx 4.6$), confirming it as the dominant effect. *Weekend share* lies furthest to the left but with a very wide CI crossing 1, indicating high uncertainty. *Temperature* and *ground-source heat pump* show clear reductions ($RR < 1$), while *floor heating*, *number of residents*, and *living area* show moderate to small increases. *Heating degree days* sits right at 1 and is the only non-significant predictor (shown in grey).

Rate Ratios with 95% Confidence Intervals



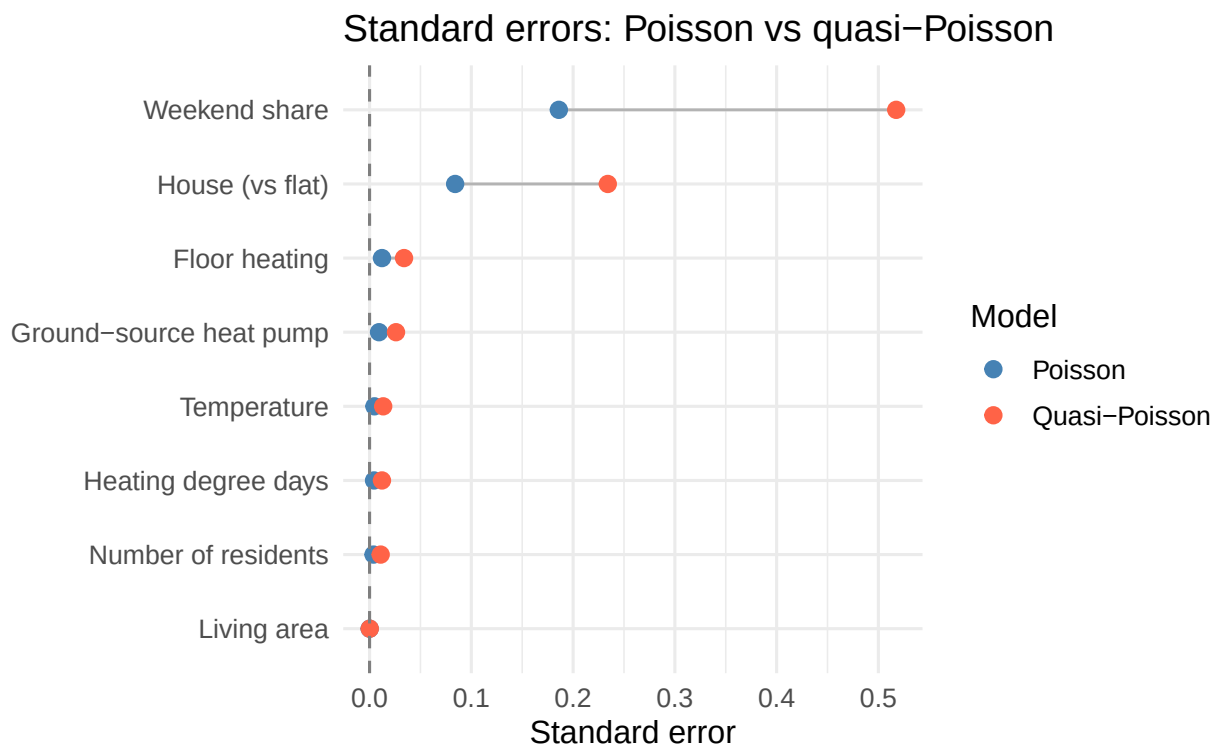
Model Diagnostics

A key assumption of the Poisson GLM is *equidispersion*, i.e., that the conditional variance equals the conditional mean. We assess this by comparing the residual deviance to the residual degrees of freedom. For the fitted Poisson model, the residual deviance is 2.75296×10^4 on 3291 df, yielding a ratio of 8.37. Since

this substantially exceeds 1, there is strong evidence of *overdispersion* — the Poisson model underestimates standard errors and overstates statistical significance.

We therefore refit the model using a *quasi-Poisson* family, which retains identical point estimates but inflates standard errors by $\sqrt{\hat{\phi}}$ ($\hat{\phi} = 7.74$), yielding more conservative and reliable inference. The most visible consequence is that *weekend share* shifts from significant under Poisson ($p = 0.028$) to clearly non-significant under quasi-Poisson ($p = 0.43$), indicating that this effect should not be over-interpreted.

The table and plot below show the SE inflation across all predictors. The effect is proportional — predictors with larger SEs to begin with (e.g., *weekend share*, *house vs flat*) show the largest absolute increase, while those with very small SEs (e.g., *living area*, *number of residents*) are inflated by a smaller absolute amount but the same factor $\sqrt{\hat{\phi}}$.

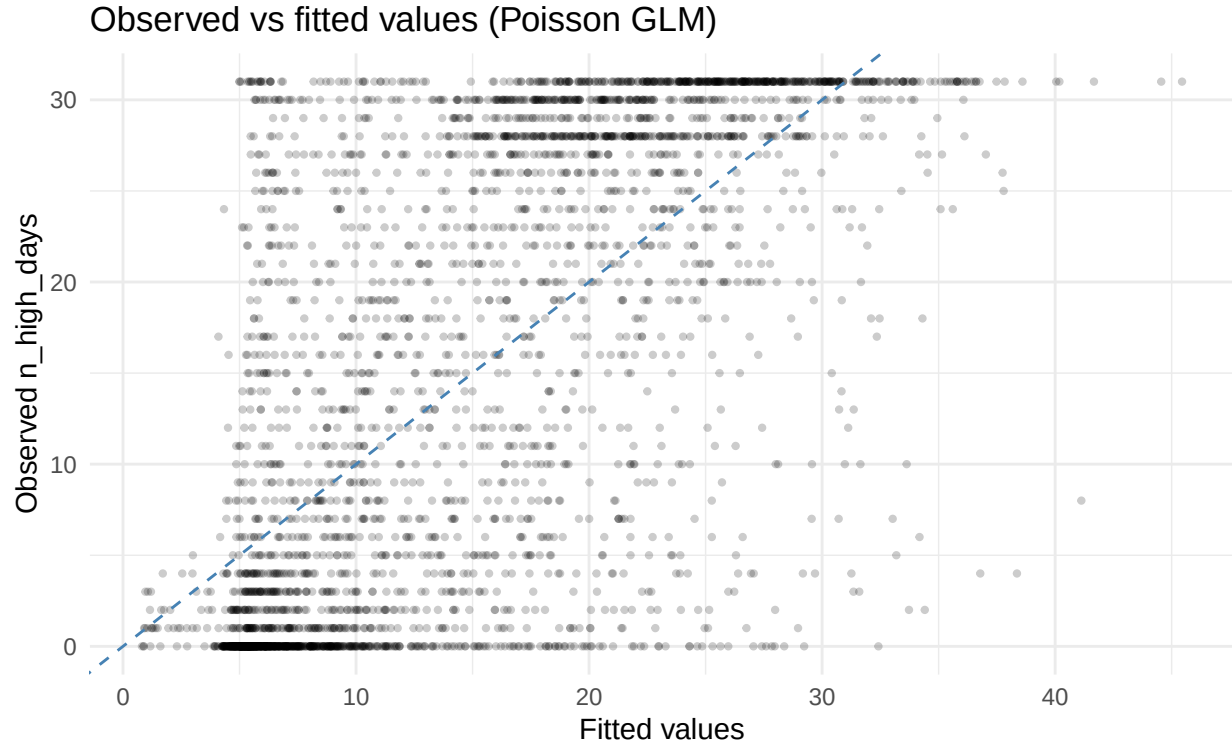


Goodness of Fit

To assess predictive accuracy, we compare fitted values against observed counts and report the Root Mean Squared Error (RMSE). The observed-vs-fitted plot is rendered as a hexbin to handle overplotting given the large number of observations.

Table 1: Root Mean Squared Error for Poisson and quasi-Poisson models.

Model	RMSE
Poisson	9.452
Quasi-Poisson	9.452



The RMSE of 9.452 days reflects the residual spread visible in the plot — predictions are reasonably centred along the identity line but with substantial scatter, partly explained by the high overdispersion ($\hat{\phi} = 7.74$). Note that the quasi-Poisson model produces identical fitted values and therefore the same RMSE, as it only affects standard errors and inference, not point predictions.

Conclusion

The Poisson GLM provides a clear and interpretable account of the factors driving the frequency of high-consumption days. Building type, temperature, and household size emerge as the strongest and most robust predictors, consistent with both domain knowledge and the exploratory analysis.

However, a dispersion ratio of 8.37 confirms substantial overdispersion, making standard Poisson inference unreliable. The quasi-Poisson model corrects for this by inflating standard errors appropriately, leading to more conservative and trustworthy conclusions — most notably, the loss of significance for weekend share.

While the quasi-Poisson model improves inference, a negative binomial model could be considered as a more flexible alternative, as it models the overdispersion through an explicit distributional parameter rather than a post-hoc correction.